

Understanding the

Maintenance Analysis Tab

A Comprehensive, Mathematical Learner's Guide

Tab 9 of app_jao_NP_API_fix_d14.py
Nordic Flow-Based Propagation — fi_no3

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Preface

This book teaches the Maintenance Analysis tab from the ground up, including the actual mathematics underneath every button — with particular depth on the hypothesis-testing machinery, since that is what "Run Analysis" and every verdict downstream of it actually rests on. It does not assume you already know what a fixed-effects estimator is, how a clustered standard error is computed, why a p-value means what it means, or what a gradient-boosted tree actually optimizes. Every idea is derived, not merely cited, and the estimation and hypothesis-testing chapters include fully worked numeric examples — real numbers, computed by hand, that you can reproduce yourself.

The tab itself is one of nine tabs in an extended tkinter application (`app_jao_NP_API_fix_d14.py`) built on a shared analytical backend (`propagation.py`). It re-runs, inside one self-contained corner of that application, the same validation the rest of the project performs: does a Nordic country's maintenance outage measurably move another bidding zone's published flow-based market parameters?

How this book is organised. Part I builds the domain vocabulary and the one governing accounting formula. Part II builds the full statistical machinery from scratch — the population regression model, the OLS derivation, the Gauss–Markov theorem, panel fixed effects and the omitted-variable-bias problem they solve, heteroskedasticity and clustered standard errors, confidence intervals, Type I/Type II error and power, and multiple-testing correction — each with a derivation and at least one fully worked numeric example. Part III is the mathematical centre of the book: each of the six hypotheses this tab tests, H1 through H6, given its own chapter — formal null and alternative hypotheses, full regression specification, worked numeric interpretation, and the specific pitfalls each one guards against — followed by chapters on economic significance, collinearity, and one complete walkthrough from a raw outage event to a final verdict. Part IV introduces counterfactual (interrupted-time-series) modeling and the one rule that governs it. Part V derives every one of the fifteen counterfactual methods. Part VI tours the tab itself, sub-tab by sub-tab. Part VII is for whoever maintains this code next. Three appendices collect notation, a formula quick-reference, and a glossary.

Notation used throughout: subscript i indexes a CNEC (a cross-sectional entity); subscript t indexes a timestamp; subscript g indexes a cluster; a hat (e.g. $\hat{\beta}$) denotes an estimated quantity; a bar (e.g. $\bar{\beta}$) denotes an average; a tilde (e.g. $\tilde{\beta}$) denotes a

within-transformed (demeaned) quantity. Read Parts I–V once, in order, with a pencil — work the examples yourself before reading the stated answer. After that, this book works equally well as a reference.

Part I — Foundations

Chapter 1. What This Tab Does, and Why It Exists

Nordic transmission system operators (TSOs) — Fingrid in Finland, Svenska kraftnät in Sweden, Statnett in Norway, Energinet in Denmark — take planned and forced maintenance outages on their own grids. An outage on one country's grid can, in principle, ripple into a completely different bidding zone's electricity market parameters, because the Nordic day-ahead market is coupled across borders through a shared capacity-allocation mechanism called Flow-Based Market Coupling (FBMC — Chapter 2 unpacks this properly).

The Maintenance Analysis tab exists to test that ripple, for one specific outage and one specific target zone, entirely inside one screen: load the market data, load the outage events, run a battery of statistical tests, drill into one event in detail, and export what you found.

Nothing in this tab — or anywhere else in this project — simulates or re-implements the FBMC capacity calculation itself. Every number here is either read directly from a published market data export, read from an outage-events feed, or a statistic computed from those two things. This is a **diagnostic and inferential tool**, built on real statistical and time-series machinery, not a market simulator.

Chapter 2. The Vocabulary You Need

FBMC replaces the older, more intuitive idea of trading capacity zone-pair by zone-pair with a different question: for every critical line in the grid, how much can every zone's net position change before that specific line's limit is reached? The market-clearing algorithm then finds the prices and volumes that respect every critical line's limit simultaneously. This is why every FBMC parameter below is defined **per critical line**, not per zone pair.

The following terms recur constantly. Read this table twice before continuing.

Term	Meaning
CNEC	Critical Network Element with Contingency — one specific grid line, under one specific contingency scenario, whose capacity limit FBMC tracks.
Bidding zone	A market area with one day-ahead electricity price (FI, NO1–5, SE1–4, DK1–2, ...).

Term	Meaning
PTDF	Power Transfer Distribution Factor — how much a CNEC's flow changes per MW of net-position change in a given zone. A linear sensitivity derived from grid topology.
Net position	A zone's total scheduled export minus import for one market time unit.
fall / F_allReference	The CNEC's reference-case flow from the day-ahead common grid model — the flow term that enters the RAM formula. Not the same quantity as fref/f0.
fref / f0	The reference flow at the CGMA net position — tracks fall numerically but is a different physical quantity, and is NOT an input to the RAM formula.
Fmax	The CNEC's thermal rating, in MW.
FRM	Flow Reliability Margin — a structural, slowly-recalibrated margin. It should not move because of one specific outage; H6 (Chapter 22) tests exactly that.
fnrao (FRA)	Flow from Non-costly Remedial Actions and Other adjustments — capacity credited back to the CNEC.
AMR	Adjustment for Minimum RAM — a floor ensuring RAM never goes negative.
FAAC (AAC)	Already Allocated Capacity — capacity pre-committed via long-term rights.
IVA	Individual Validation Adjustment — TSO-discretionary, almost always zero.
RAM	Remaining Available Margin — the capacity left for day-ahead trading on a CNEC. See Chapter 3.
Shadow price	The marginal value (currency per MW) of relaxing a CNEC's binding constraint by one more MW. Zero when not binding.
MTU	Market Time Unit — 15 minutes, the native resolution of the data.

Two formulas anchor the rest of this book.

$$RAM = Fmax - FRM - fall + fnrao + AMR - FAAC - IVA$$

$$flow_{CNEC} \approx \sum_z PTDF_z \times NetPosition_z$$

Chapter 3. The RAM Identity — The Formula Underneath Everything

The first formula above is an accounting identity, not an approximation. Every term is independently published in the market data, and their signed sum equals the published RAM column to within about one megawatt on real data. This tab checks that balance every time it builds its covariates.

Formally, writing the seven terms as $X_1, \dots, X_7$ with signs $s_1, \dots, s_7 \in \{+1, -1\}$:

$$RAM_t = \sum_{k=1}^7 s_k \cdot X_{k,t}, \text{ with } s = (+1, -1, -1, +1, +1, -1, -1)$$

for (Fmax, FRM, fall, fnrao, AMR, FAAC, IVA) in that order. An earlier version of this exact formula was wrong — it silently dropped the fnrao term, producing RAM values 300–400 MW off on real data, and the only check of that error at the time was circular. Every regression in Part III, and one whole physics-informed forecasting method (Chapter 34), rests on this identity being complete and correctly signed.

Part II — The Mathematics of Estimation

This part builds, from nothing, every piece of statistical machinery Part III leans on. Nothing here is specific to power grids — it is the general theory of estimating a treatment effect from observational panel data, worked with actual numbers so each formula has a face.

Chapter 4. The Population Regression Function and the Assumptions Behind It

Start from the question itself: does some outage-related regressor x cause a change in some outcome y ? The **population regression function** posits a true, unobserved relationship

$$y_i = \alpha + \beta x_i + \varepsilon_i$$

where α and β are fixed but unknown population parameters, and ε_i is the part of y_i not explained by x_i — every other cause, however small. β is the object of interest: the population-average change in y for a one-unit change in x . Nothing about this equation is yet estimable; it describes the world, not a procedure for learning about it from a sample.

Four assumptions make β learnable from data, and are worth stating explicitly because every later refinement in this part exists to fix a violation of one of them:

- **Linearity in parameters.** The *relationship being modeled* is linear in α and β (x itself may be transformed — log, squared, a dummy — before it enters the model).
- **Exogeneity.** $E[\varepsilon_i | x_i] = 0$ — the regressor carries no information about the average error. Violate this (e.g., if outages are more likely exactly when RAM is already unusually low, for reasons unrelated to the outage itself) and β is systematically biased, not just noisy.
- **No perfect collinearity.** No regressor is an exact linear function of the others — otherwise $(X'X)$ in Chapter 5 cannot be inverted at all. Chapter 24 covers the near-collinear case this project actually encounters.
- **Independent, identically distributed sampling** (relaxed substantially in Chapters 9 and 13, which is most of what makes a panel model different from the simple case here).

Chapters 5–16 build up exactly the machinery needed when these assumptions hold only partially — first deriving the ideal case, then relaxing it one assumption at a time until it matches what this tab's real data actually looks like.

Chapter 5. Deriving the OLS Estimator

Given a sample of n observations (x_i, y_i) , **Ordinary Least Squares** chooses the values α, β that make the fitted line come as close as possible to the data, in the specific sense of minimising the sum of squared vertical distances (residuals):

$$S(\alpha, \beta) = \sum_{i=1}^n (y_i - \alpha - \beta x_i)^2$$

Squared, rather than absolute, error is chosen deliberately: S is then a smooth, differentiable, strictly convex function of (α, β) , guaranteeing a unique minimum reachable by calculus alone. Differentiate with respect to each parameter and set both to zero (the **normal equations**):

$$\begin{aligned} \partial S / \partial \alpha &= -2 \sum (y_i - \alpha - \beta x_i) = 0 \rightarrow \sum y_i = n\alpha + \beta \sum x_i \\ \partial S / \partial \beta &= -2 \sum x_i (y_i - \alpha - \beta x_i) = 0 \rightarrow \sum x_i y_i = \alpha \sum x_i + \beta \sum x_i^2 \end{aligned}$$

Solving this 2×2 linear system for β gives the familiar covariance-over-variance form,

$$\beta = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \text{Cov}(x, y) / \text{Var}(x)$$

With more than one regressor, stack a column of ones and every x -variable into a design matrix X ($n \times k$) and y into an n -vector, and the identical minimisation (now over a vector β) yields, in matrix form,

$$\beta = (X'X)^{-1} X'y$$

This single expression is the workhorse of the entire codebase's statistical layer: every panel regression in Part III and the closed-form Ridge method in Chapter 32 are variants of exactly this formula, differing only in what X contains and what is added to $(X'X)$ before inverting it.

Chapter 6. A Fully Worked OLS Example, By Hand

Six observations of a RAM-like outcome, three before an outage and three during it:

Row i	x_i (outage)	y_i (RAM)	y_i – predicted
1	0	102	
2	0	98	
3	0	101	

Row i	x _i (outage)	y _i (RAM)	y _i – predicted
4	1	95	
5	1	93	
6	1	97	

Following Chapter 5's formula with $x = 0.5$, $y = 97.667$:

$$\beta = \text{Cov}(x,y)/\text{Var}(x) = -5.333, \alpha = \bar{y} - \beta \bar{x} = 100.333$$

So the fitted line is $\hat{y} = 100.333 - 5.333 \cdot x$ — predicting 100.333 with no outage, 95.0 with one. The six residuals ($y_i - \hat{y}_i$) are 1.667, -2.333, 0.667, 0, -2, 2, giving a residual sum of squares $SSE=16.667$. With $n=6$ observations and 2 estimated parameters (α, β), there are $n-2=4$ residual degrees of freedom, so the estimated error variance and the standard error of β are

$$\sigma^2 = SSE/(n-2) = 4.167, SE(\beta) = \sqrt{\sigma^2 / \sum (x_i - \bar{x})^2} = 1.667$$

$$t = \beta / SE(\beta) = -3.20, p = 0.033 \text{ (t-distribution, } df=4\text{)}$$

$p=0.033$ clears the conventional 0.05 threshold: at this (tiny, illustrative) sample size, the -5.33 MW effect is statistically distinguishable from zero. Chapter 15 returns to exactly how small a sample like this can be before that conclusion becomes unreliable.

Chapter 7. The Gauss–Markov Theorem: Why OLS Is BLUE

Why square the errors at all, rather than some other penalty? Under the four assumptions of Chapter 4 plus one more — **homoskedasticity**, $\text{Var}(\varepsilon_i) = \sigma^2$ for every i , the error's spread doesn't itself depend on x_i — the **Gauss–Markov theorem** states that the OLS estimator is BLUE: the Best (minimum-variance) Linear Unbiased Estimator. “Linear” means β is a linear function of the y 's; “unbiased” means $E[\hat{\beta}] = \beta$, it is centred on the truth; “best” means no other linear unbiased estimator has smaller variance, for any true β . This is why OLS, rather than some other fitting rule, is the default: under these assumptions it is provably the most precise unbiased choice available, not merely a convenient one.

The qualifier matters as much as the theorem. Real CNEC data is not homoskedastic — a CNEC that rarely binds has far less shadow-price variance than one that binds often — and observations sharing a timestamp are not independent (Chapter 4's fourth assumption). Chapters 9–13 relax exactly these two assumptions, in order: fixed effects

first (a structured violation of independence across a panel's own entities), then robust and clustered standard errors (which keep β itself unchanged — it is still BLUE under the weaker assumptions that remain — but replace the *variance formula* used to judge its precision).

Chapter 8. Goodness of Fit: R^2 , Within R^2 , and Overall R^2

R^2 measures the fraction of y's variation the fitted model explains:

$$R^2 = 1 - SSE/SST, SST = \sum (y_i - \bar{y})^2$$

In Chapter 6's example, $SST = \sum (y_i - \bar{y})^2 = 53.33$, so $R^2 = 1 - 16.667/53.33 = 0.687$ — the outage dummy explains about 69% of this tiny sample's RAM variation. Once entity fixed effects (Chapter 9) enter the picture, two different R^2 figures become meaningful and this tab reports both: **R^2_{within}** , computed on the demeaned data actually used to identify β (how well the model explains variation *within* each CNEC over time, the only variation fixed effects can possibly explain), and **R^2_{overall}** , computed on the raw, un-demeaned data (inflated by however much of y's total variance is just cross-CNEC baseline differences the entity intercepts soak up for free). R^2_{overall} is almost always the larger, more flattering number; R^2_{within} is the honest one for judging whether the *outage variable itself* explains much.

Chapter 9. Panel Data and the Fixed-Effects (Within) Estimator

This tab's data isn't one cross-section — it's a **panel**: many CNECs (indexed i), each observed repeatedly over time (indexed t). Pooling every (i,t) row into one plain OLS regression like Chapter 5's implicitly assumes every CNEC shares the same baseline level, which is false, and Chapter 10 shows concretely how wrong ignoring that can be.

The **fixed-effects** model adds one intercept per entity:

$$y_{it} = \alpha_i + \beta D_{it} + \varepsilon_{it}$$

where D_{it} is the outage-active dummy (or dose) and α_i absorbs every CNEC's own time-invariant baseline. Estimating every α_i directly for hundreds of CNECs is wasteful; the **within transformation** instead demeans every variable by its own entity's average across time,

$$\bar{y}_i = y_{it} - \bar{y}_i, D_{it} = D_{it} - \bar{D}_i$$

and applies plain OLS (Chapter 5) to the demeaned data: $\beta = (D'D)^{-1}D'y$. This is mathematically identical to including a full set of CNEC dummy variables in one giant regression (Least Squares Dummy Variables, LSDV) but is computed far more cheaply — subtracting each entity's own mean achieves exactly the same thing as estimating and then netting out its own dummy coefficient. Because it removes anything constant *within* a CNEC over the sample window, fixed effects answer exactly the right question: does THIS CNEC's own value move when the outage variable turns on, never are CNECs with more outages simply different CNECs.

Chapter 10. A Fully Worked Fixed-Effects Example: Omitted-Variable Bias, Made Concrete

Two CNECs, deliberately unbalanced: CNEC A (high baseline RAM) is observed mostly *before* an outage; CNEC B (low baseline RAM) is observed mostly *during* one.

i	t	x_it (outage)	y_it (RAM)
A	1–4	0,0,0,0	200,198,202,199
A	5	1	190
B	1	0	150
B	2–5	1,1,1,1	143,147,144,146

Pooled OLS (Chapter 5, ignoring which CNEC each row belongs to) gives $\beta_{\text{pooled}} = -35.8$ — a huge, alarming effect. But it is badly wrong: it is partly measuring the outage, and partly just measuring "CNEC B has a ~50 MW lower baseline than CNEC A," which has nothing to do with any outage — the outage-active rows happen to be disproportionately CNEC B's rows, so the entity difference leaks into β through the back door. This is **omitted-variable bias**: excluding a relevant grouping variable (which CNEC) that is correlated with both y and the included regressor.

Fixed effects (Chapter 9) removes it: demeaning each CNEC's own observations by its own mean first,

$$A: \bar{y}_A = 197.8 \quad B: \bar{y}_B = 146.0$$

$$\beta_{FE} = -7.375, SE=1.575, t=-4.68, p=0.0023 \quad (df = n - \text{entities} - 1 = 10 - 2 - 1 = 7)$$

The honest, entity-adjusted effect is -7.4 MW, not -35.8 — still real and still statistically significant, but roughly a fifth the size the naive pooled regression implied. This single

worked contrast is the entire justification for Chapter 5's fixed-effects specification appearing in every one of Part III's six hypotheses: pooling across CNECs without entity intercepts does not merely add noise, it can systematically manufacture an effect several times too large.

Chapter 11. The Frisch–Waugh–Lovell Theorem: What “Controlling For” Really Means

Chapter 9 asserted, without proof, that demeaning is equivalent to including a full set of entity dummies. The **Frisch–Waugh–Lovell (FWL) theorem** is the general statement behind that claim, and behind the whole idea of “controlling for” a variable in any regression. Suppose the model is $y = X\beta + Z\gamma + \varepsilon$, where X is the regressor of interest and Z is a set of control variables (here, entity dummies). FWL says β from the full regression can be recovered in three steps: (1) regress y on Z alone and keep the residuals $\hat{u}$; (2) regress X on Z alone and keep the residuals $\hat{X}$; (3) regress $\hat{u}$ on $\hat{X}$. The resulting slope is *exactly* β from the full multi-variable regression.

Applied to fixed effects: regressing y_{it} on the entity dummies alone and taking residuals is precisely the within-transformation $y_{it} - \bar{y}_i = y_{it} - \bar{y}_i$ from Chapter 9 (each entity dummy's own OLS fit is just that entity's mean); the same for D_{it} . Step 3 is then Chapter 9's demeaned regression. FWL is the rigorous reason “control for CNEC” and “demean by CNEC” are the same operation, not merely a computational shortcut that happens to give a similar answer — and it generalises to controlling for anything (a set of calendar-month dummies, a linear trend), which is why it is worth knowing as a general tool, not just as this project's own fixed-effects trick.

Chapter 12. Heteroskedasticity and the Robust (White) Standard Error

Gauss–Markov's variance formula, $\sigma^2(X'X)^{-1}$, assumed every residual has the *same* variance σ^2 . **Heteroskedasticity** — unequal error variance across observations — is the norm here, not the exception: a CNEC that rarely binds has shadow price pinned near zero with little variance; one that binds often has much more. Under heteroskedasticity, OLS's β remains unbiased (Chapter 5's derivation never used homoskedasticity), but the *classical* variance formula becomes wrong, typically understating the true uncertainty.

White's **heteroskedasticity-robust** standard error replaces the single σ^2 with each observation's own squared residual, $u_i^2 = (y_i - \hat{y}_i)^2$, inside a sandwich-shaped estimator:

$$Var_{robust}(\beta) = (X'X)^{-1} \left[\sum_i u_i^2 x_i x_i' \right] (X'X)^{-1}$$

This is consistent (converges to the true variance as n grows) whatever the actual pattern of heteroskedasticity turns out to be, without needing to specify that pattern in advance. It is, however, still built on an assumption Chapter 13 shows this data violates too: that different observations' errors are independent of each other. Chapter 13's clustered SE is the next, and for this tab's data the more important, relaxation.

Chapter 13. Clustered Standard Errors, Derived and Worked

Observations from the same time period are not independent: a shock at 14:00 on one date plausibly affects several CNECs simultaneously (shared weather, a shared market condition), so errors from *different* CNECs at the *same timestamp* are correlated, even though errors from *different timestamps* are more nearly independent. Treating correlated observations as independent — as both Chapters 7 and 12's formulas do — understates the true uncertainty, inflating the apparent significance of β .

Cluster-robust standard errors only assume independence *across* clusters (here, across timestamps), not within them. Grouping observations into clusters $g=1, \dots, G$, the clustered sandwich estimator is

$$Var_{cluster}(\beta) = (X'X)^{-1} \left[\sum_{g=1}^G X_g' u_g u_g' X_g \right] (X'X)^{-1}$$

where X_g and u_g stack the regressors and residuals for every observation in cluster g . Chapter 12's robust formula is the special case where every cluster has exactly one member ($u_g u_g'$ collapses to a single u_i^2); clustering is strictly more general, since it also captures the *covariance* between different members of the same cluster's errors, not just each one's own variance.

Worked illustration. Simulate twelve fixed-effects-demeaned residuals: six CNECs observed at the same six timestamps, with a shared shock at each timestamp (three ordinary and three sharply negative "during-outage" timestamps, identical across all six CNECs that share them) on top of small idiosyncratic noise:

$$SE_{naive (unclustered)} = 0.151, SE_{time\text{-}clustered} = 0.262 \text{ (73\% larger)}$$

$$t_{naive} = -51.6 \text{ vs. } t_{clustered} = -29.8$$

Both are still significant in this deliberately dramatic illustration, but the gap shows the mechanism plainly: because the six CNECs' errors move together at each timestamp rather than independently, the naive formula effectively — and wrongly — treats six correlated observations as six independent ones, understating the true sampling uncertainty by nearly half. In a less extreme, more realistic case the same mechanism can move a result from significant to not. `run_panel_regression()` requests time-clustered standard errors by default for exactly this reason, and records a documented fallback chain (time-clustered → entity-clustered → robust → unadjusted classical) whenever the requested mode cannot be estimated — each step relaxing to a weaker, more clearly-stated assumption than the last, with every result recording honestly which one it actually rested on.

Chapter 14. Confidence Intervals and the Duality With Hypothesis Tests

A point estimate β alone hides how much sampling variability it carries. A **confidence interval** expresses that variability directly, as a range of values plausible given the data:

$$\beta \pm t_{\alpha/2, df} \cdot SE(\beta)$$

where $t_{\alpha/2, df}$ is the critical value cutting off $\alpha/2$ in each tail of the t-distribution with the model's residual degrees of freedom (for large df this converges to the familiar ± 1.96 for a 95% interval). Applying this to Chapter 6's example ($\beta = -5.333$, $SE = 1.667$, $df = 4$, $t_{0.025, 4} \approx 2.776$) gives a 95% interval of roughly $[-10.96, +0.29]$ — note it is wide enough here to still barely contain zero at the 95% level despite $p = 0.033$ clearing 0.05, a reminder that the exact boundary depends on which reference distribution and rounding are used, not a contradiction.

The **duality** worth internalising: a 95%-confidence interval that excludes zero is mathematically equivalent to a two-sided test rejecting $H_0: \beta = 0$ at the 5% level, and vice versa. A confidence interval is simply the set of every hypothesised value that a hypothesis test, run against that specific value, would fail to reject. This is why this tab's regression output (Chapter 17 onward) can be read either way — as a p-value crossing 0.05, or as an interval that does or doesn't contain zero — and both readings always agree.

Chapter 15. Type I and Type II Error, Power, and Sample Size

Every hypothesis test can go wrong in exactly two ways, and the two are in tension with each other:

	H ₀ actually true	H ₀ actually false
Test rejects H ₀	Type I error (rate α)	Correct rejection (power)
Test fails to reject	Correct (rate $1-\alpha$)	Type II error (rate β)

Type I error is a false positive — concluding an outage moved a parameter when it really didn't — controlled directly by the significance threshold α (0.05 by convention, tightened further by Holm–Bonferroni in Chapter 16). **Type II error** is a false negative — missing a real effect — and **power** ($1-\beta$) is the probability of correctly detecting a real effect of a given size. Power depends on three things: the true effect size (bigger effects are easier to detect), the noise level σ (noisier data hides smaller effects), and the sample size n (SE shrinks roughly as $1/\sqrt{n}$, so quadrupling the sample halves the standard error).

This has a direct, practical consequence for this tab: a short pre-period or few outage episodes means low power, and a "not significant" H1–H4 verdict on thin data is genuinely ambiguous between "no real effect" and "too little data to detect the effect that's actually there" — it should never be read as positive evidence of **no** effect, only as an absence of evidence **for** one. H5's tiny worked logit example in Chapter 21 ($\gamma_{H5}=2.77$, a large apparent effect, yet $p=0.080$, not significant at $n=10$) is exactly this phenomenon made concrete: the effect size is real and large in the sample, but the sample is simply too small for that size to clear the significance bar with confidence.

Chapter 16. Multiple Testing: Family-Wise Error Rate and the Holm–Bonferroni Procedure, Worked

Run one test at $\alpha=0.05$ and the false-positive rate is, by construction, 5%. Run m **independent** tests and ask for the probability that **at least one** falsely rejects — the **family-wise error rate (FWER)** — and it climbs:

$$FWER = 1 - (1-\alpha)^m$$

For $m=4$ (H1–H4, tested on the same dataset), $FWER = 1-0.95^4 \approx 18.5\%$, nearly four times the nominal 5% — even if every single null hypothesis is exactly true, there is close to a one-in-five chance **some** test shows a spurious significant result purely from

running four of them.

The (unmodified) **Bonferroni** correction controls this crudely, by testing every hypothesis against α/m instead of α . **Holm–Bonferroni** is uniformly more powerful (rejects at least as much, often more) while still controlling the same FWER: sort the m p-values ascending, $p_{(1)} \leq \dots \leq p_{(m)}$, and step through them from smallest to largest, rejecting each as long as

$$p_{(k)} \leq \alpha / (m - k + 1)$$

stopping at the first k where this fails — every hypothesis from that point on is left un-rejected, regardless of whether a *later*, larger p-value would individually clear *its own* threshold in isolation. A concrete worked example, four hypotheses at $\alpha=0.05$:

k	Hyp.	p_(k) vs. $\alpha/(m-k+1)$	Verdict
1	H1	0.001 vs. $0.05/4 = 0.0125$	reject
2	H4	0.020 vs. $0.05/3 = 0.0167$	fail — stop here
3	H3	0.030 vs. $0.05/2 = 0.0250$	not rejected (procedure already stopped)
4	H2	0.040 vs. $0.05/1 = 0.0500$	not rejected (procedure already stopped)

The easy mistake: H2's own raw p-value (0.040) is below its own row's threshold (0.05) and below α itself — it would be "significant" tested alone — but because the step-down procedure already failed at $k=2$, H2 is correctly reported as not significant under the family-wise correction. `summarize_hypotheses()` applies exactly this step-down logic across H1–H4 before labelling any of them "SUPPORTED."

Part III — The Six Hypotheses, In Full

Every regression below is an instance of Chapter 9's fixed-effects model, fit with Chapter 13's clustered standard errors, read through Chapters 14–16's testing framework. This part gives each hypothesis its own chapter: the formal null and alternative, the exact specification, what a verdict means substantively, and the pitfall it specifically guards against.

Chapter 17. H1 in Full: HVDC Outages and the Reference Flow

$$H_0: \beta_{\text{HVDC}} = 0 \text{ vs. } H_A: \beta_{\text{HVDC}} \neq 0$$

$$\text{fall}_{it} = \alpha_i + \beta_1 \cdot \text{HVDCOutage}_{it} + \varepsilon_{it}, \text{ clustered by } t$$

`fall` is the sign-normalised reference flow (“fall_signed,” falling back to raw `fall` if unavailable), constructed so a source-country HVDC outage's effect has a *consistent expected sign* regardless of any individual CNEC's own flow-direction convention — without this normalisation, genuinely identical physical effects on differently-oriented CNECs could cancel each other out in the pooled panel rather than reinforcing. H1 is supported only when β_{HVDC} clears both Holm–Bonferroni-corrected $p < 0.05$ (Chapter 16) *and* the economic-significance floor for flow (Chapter 23) — a large, precisely-estimated, but economically trivial β_{HVDC} is reported as merely statistically significant, never as “supported.”

Chapter 18. H2 in Full: AC Line Outages and PTDF Sensitivity

$$H_0: \beta_{\text{AC}} = 0 \text{ vs. } H_A: \beta_{\text{AC}} \neq 0$$

$$|\text{PTDF}_{\text{SRC}}|_{it} = \alpha_i + \beta_2 \cdot \text{ACOutage}_{it} + \varepsilon_{it}, \text{ clustered by } t$$

Using the **absolute** PTDF value is the specification choice that makes this a clean test of a pure topology effect: an AC line outage changes the grid's admittance matrix, which changes how *sensitive* a CNEC is to the source zone's net position — the sensitivity can grow or shrink depending on which lines are still available, but $|\text{PTDF}|$ growing either way is the signature of a topology change, independent of how much power happens to be flowing at any given moment (which is what H1's flow-level test already covers). Fitting the raw signed PTDF instead would risk a topology-driven increase in one direction cancelling a topology-driven decrease in another, purely from sign, even though both are the same underlying phenomenon.

Chapter 19. H3 in Full: Forced Outages and RAM

$$H_{\blacksquare}: \beta_{\blacksquare} = 0 \text{ vs. } H_{\blacksquare}: \beta_{\blacksquare} \neq 0$$

$$RAM_{it} = \alpha_i + \beta_3 \cdot ForcedOutage_{it} + \varepsilon_{it}, \text{ clustered by } t$$

Because RAM is itself Chapter 3's seven-term identity, a significant $\beta_{\blacksquare}$ says the *combined*, already-netted effect across every term moved — it does not by itself say which term(s) moved or by how much. That attribution is a separate, retrospective calculation (the Δ RAM decomposition pane, Chapter 39), applying the same identity to one specific event's before/during averages rather than to a population-level regression coefficient. H3's role is purely to establish *whether* a combined movement exists at all, at the population level, before drilling into any one event's own breakdown.

Chapter 20. H4 in Full: Forced Outages and the Shadow Price

$$H_{\blacksquare}: \beta_{\blacksquare} = 0 \text{ vs. } H_{\blacksquare}: \beta_{\blacksquare} \neq 0$$

$$shadowPrice_{it} = \alpha_i + \beta_4 \cdot ForcedOutage_{it} + \varepsilon_{it}, \text{ fit only where } shadowPrice_{it} > 0, \text{ clustered by } t$$

H4 is fit on the **binding subsample** only — rows where the CNEC actually has a positive shadow price — since a non-binding CNEC's shadow price is exactly zero *by construction* (Chapter 2), not because nothing happened; including those structural zeros would mix two different questions, "did the outage make this constraint bind harder" and "was this CNEC ever going to bind regardless," into one coefficient that answers neither cleanly. If fewer than fifty binding rows exist in the loaded window, H4 is skipped outright rather than fit on too little data (see Chapter 15's power discussion) to trust the result at all.

Chapter 21. H5 in Full: The Logistic Regression for IVA, Derived and Worked

IVA is a discretionary correction, usually exactly zero, so H5 cannot be a linear regression on IVA's *size* the way H1–H4 are on their outcomes — it is a test of whether IVA activates *at all*, more often under forced than planned outages. Writing $p_{\blacksquare} = P(IVA_{it} > 0)$, the **logit model** relates the *log-odds* of activation linearly to the regressor:

$$H_{\blacksquare}: \gamma_{\blacksquare} = 0 \text{ vs. } H_{\blacksquare}: \gamma_{\blacksquare} > 0$$

$$\ln(p_{it} / (1 - p_{it})) = \gamma_0 + \gamma_1 \cdot Forced_{it}$$

Unlike every model in this book so far, this has no closed-form solution — γ is found by **maximum-likelihood estimation**, numerically maximising the probability the model assigns to the actually-observed 0/1 outcomes over choices of γ .

Worked example. Ten outage events, five planned and five forced, with IVA activation recorded 0/1:

Type	n	IVA active	Sample rate
Planned	5	1 of 5	0.20
Forced	5	4 of 5	0.80

Maximum-likelihood estimation on this saturated two-group model recovers the log-odds exactly matching each group's own observed rate:

$$\gamma_0 = \ln(0.20/0.80) = -1.386, \gamma_1 = \ln(0.80/0.20) - \gamma_0 = 2.773$$

confirming $P(\text{IVA}|\text{planned})=0.20$, $P(\text{IVA}|\text{forced})=0.80$ exactly. For a binary regressor comparing two proportions, the standard error of γ has a direct closed form (from the delta method applied to each group's own binomial variance),

$$SE(\gamma_1) = \sqrt{[1/(n_1 p_1 (1-p_1)) + 1/(n_0 p_0 (1-p_0))]} = 1.581$$

$$z = \gamma_1 / SE(\gamma_1) = 1.75, p = 0.080 \text{ (two-sided, normal reference)}$$

A large, real-looking effect (four-in-five versus one-in-five) that nonetheless narrowly misses conventional significance at $n=10$ — precisely Chapter 15's power lesson, now worked through H5's own actual test rather than in the abstract. H5's real data, with far more than ten events pooled across a full sample window, has correspondingly more power than this illustration; the mechanism is identical either way.

Chapter 22. H6 in Full: The Placebo Test

$H_0: \beta = 0$ vs. $H_1: \beta \neq 0$ (the project WANTS to fail to reject H_0 here)

$$FRM_{it} = \alpha_i + \beta_6 \cdot \text{Outage}_{it} + \varepsilon_{it}, \text{ clustered by } t$$

Same specification family as H1–H4, but fit on a variable that is structural (Chapter 2) and should be inert to any single outage. H6 inverts the usual reading: finding β significant is evidence *against* the estimator, not evidence for a real FRM effect — it means whatever machinery produced H1–H4's other verdicts is capable of manufacturing a signal on this specific dataset that shouldn't be there, and every other verdict in the same

report is demoted accordingly rather than taken at face value. This is the project's own falsifiability check on itself, not merely one more line in a results table.

Chapter 23. Economic Significance: Thresholds and the Decision Table

A coefficient can be statistically significant — reliably distinguishable from zero, per Chapters 14–16 — and still be economically irrelevant: a rock-solid, $p < 0.0001$ shift of half a megawatt in a thousand-megawatt RAM budget is not something anyone would act on. **ECONOMIC_SIGNIFICANCE_THRESHOLDS** sets a conservative, documented minimum-magnitude floor per dependent variable, checked as a wholly separate bar from the p-value:

	$ \beta $ below floor	$ \beta \geq \text{floor}$
$p \geq 0.05$ (corrected)	Not significant	Not significant
$p < 0.05$ (corrected)	Statistically significant only	SUPPORTED

Only the bottom-right cell is reported as genuinely "supported"; the bottom-left cell — a precisely-estimated but tiny effect — is deliberately labelled more cautiously, so a reader scanning verdicts cannot mistake statistical precision for practical importance. These floors are calibrated defaults, not measurements against live market data (none is committed to this repository) — worth revisiting if a better-grounded number becomes available, per Chapter 27's known limitations.

Chapter 24. Collinearity Between the Dummy and the Dose

Each outage type contributes both a binary "active" dummy and a continuous "MW lost" dose variable to the regressors Chapters 17–22 can use. With few independent outage episodes, these two can become **near-perfectly collinear** — every time the dummy is 1, the dose takes nearly the same one value, so the design matrix ($X'X$) from Chapter 5 becomes close to singular (recall Chapter 4's fourth assumption, no perfect collinearity, being violated in the near-exact case rather than the exact one). Symptom: the two coefficients become individually unstable and hard to interpret, even though *jointly* they fit the data just as well as either alone would.

`_prune_collinear_dose_pairs()` runs before every fit and drops the binary duplicate, keeping the dose variable when the two are this closely tied (falling back to a deterministic rule when neither or both candidates are themselves binary) — rather than let OLS arbitrarily split one real effect's coefficient between two variables carrying almost the same information, which is what an uncorrected near-singular ($X'X$) does in practice: small changes in the data can swing the split wildly without changing the *combined* fit at all. This applies to any regressor list passed to `run_panel_regression()`, not just the built-in H1–H6 specifications.

Chapter 25. From Raw Outage to Verdict: A Complete Worked Walkthrough

Tying Chapters 5–24 together, start to finish, for one hypothesis:

- **Raw data.** An outage event (start/end, asset type, planned/forced) and a JAO export with one row per (CNEC, 15-minute MTU).
- **Covariates** (`build_covariates()`). The outage's interval is turned into a 0/1 “active” dummy aligned to every MTU, plus a continuous MW-lost dose where derivable; Chapter 3's RAM identity is checked to balance; Chapter 24's collinearity pruning runs.
- **Specification.** The right hypothesis's equation is selected (Chapters 17–22) — dependent variable, regressor, and for H4 specifically, the binding-only subsample.
- **Estimation.** Chapter 9's within-transformation removes each CNEC's own baseline; Chapter 5's OLS normal equations are solved on the demeaned data to get β .
- **Inference.** Chapter 13's clustered sandwich formula (falling back per Chapter 13's documented chain if it cannot be estimated) gives $SE(\beta)$; Chapter 14 turns that into a t-statistic and p-value.
- **Correction.** H1–H4's four p-values are jointly adjusted by Chapter 16's Holm–Bonferroni step-down procedure.
- **Economic check.** Chapter 23's magnitude floor is applied as a second, independent bar.
- **Verdict.** Only a coefficient clearing *both* bars, on a report whose H6 placebo (Chapter 22) itself did NOT reject, is reported as “SUPPORTED” — anything else is reported honestly as statistically significant only, inconclusive, or not significant, each a substantively different, clearly labelled outcome, never collapsed into one generic “significant/not significant” bit.

Part IV — Counterfactual Modeling: Foundations

Chapter 26. What “Counterfactual” Means, and the One Rule

Part III's regressions test whether outages *in general* move a parameter, pooling many events. “Single Event” asks a sharper question about *one specific outage*: what would this CNEC's series have looked like if the outage had never happened? Write the observed series as y_t , and let $t=0$ mark the outage's start. The **counterfactual** $Y_{\blacksquare}(0)_t$ (“Y-nought,” borrowing Rubin's potential-outcomes notation) is an estimate of what y_t would have been in the untreated world, fit using only pre-period data ($t<0$) and projected forward. The estimated **impact** at any moment is simply

$$impact_t = y_t - Y_{\blacksquare}(0)_t$$

Every method in Part V is a different way of constructing $Y_{\blacksquare}(0)_t$. All fifteen exist in service of one rule, worth memorising before anything else here:

A counterfactual projection may only ever be fit on data strictly before $t=0$. It must never read a real observed value from $t \geq 0$ — not directly, and not indirectly through a feature (such as “yesterday's value at this hour”) that happens to point into that window.

The indirect case is the subtle one: a naive lag feature, computed the ordinary way, would for any timestamp shortly after $t=0$ quietly read real, already-outage-affected data as its own input. Every lag-based method here instead projects the $t \geq 0$ window **recursively** — one step at a time, in chronological order, feeding each step's own prediction back in as the next step's lag source — so it only ever sees its own forecasts past $t=0$, never contaminated real data.

Part V — The Method Catalog, Mathematically

Chapter 27. Deterministic Lookup: Seasonal-Naive and Hurdle

Seasonal-naive. The simplest possible $Y_{\blacksquare}(0)$: the pre-period average for each (hour, weekday) cell. Writing $h(t)$, $d(t)$ for the hour and weekday of timestamp t , and P for the set of pre-period timestamps:

$$Y_{\blacksquare}(0)_t = (\sum_{s \in P : h(s)=h(t), d(s)=d(t)} y_s) / |\{s \in P : h(s)=h(t), d(s)=d(t)\}|$$

It has zero estimation variance beyond the cell average itself, needs no fitting procedure, and is the baseline every other method is implicitly judged against.

Hurdle. For a column with a genuine point mass at zero — shadow price is the canonical case — a smooth continuous method risks projecting a small positive baseline the real quantity structurally cannot take. The hurdle model fits *two* lookup tables on the same cells: the probability of a nonzero value, and the expected size given that it is nonzero,

$$p_{\blacksquare}(h,d) = P(y>0 \mid \text{hour}=h, \text{weekday}=d), \mu_{\blacksquare}(h,d) = E[y \mid y>0, \text{hour}=h, \text{weekday}=d]$$

$$Y_{\blacksquare}(0)_t = p_{\blacksquare}(h(t), d(t)) \times \mu_{\blacksquare}(h(t), d(t))$$

On a column that is *not* zero-inflated, $p_{\blacksquare}$ saturates near 1 and the method degrades harmlessly to plain seasonal-naive.

Worked illustration. Fifteen pre-period shadow-price observations for one (hour, weekday) cell, mostly zero with three binding spikes:

$$\text{values} = 0,0,0,0,18.2,0,0,22.5,0,0,0,15.9,0,0,0$$

$$p_{\blacksquare} = 3/15 = 0.20, \mu_{\blacksquare} = (18.2+22.5+15.9)/3 = 18.87, \blacksquare(0) = 0.20 \times 18.87 = 3.77$$

A plain seasonal-naive lookup on this same cell would instead project the raw mean of all fifteen values (also 3.77, coincidentally identical here since seasonal-naive's mean *is* $p_{\blacksquare} \times \mu_{\blacksquare}$ by definition — the two formulas are algebraically the same quantity, computed two different ways) as a *constant* baseline for every future occurrence of this cell, never zero and never a spike — exactly the "small positive baseline the real quantity cannot take" problem hurdle exists to avoid representing as if it were a typical, everyday value.

Chapter 28. Harmonic Regression and STL Decomposition

Fourier + linear trend. A smooth parametric curve: a linear trend plus K pairs of sine/cosine harmonics at the data's natural period T, estimated by Chapter 5's OLS applied to this particular design matrix:

$$y_t = \beta_0 + \beta_1 t + \sum_{k=1}^K [a_k \sin(2\pi k t / T) + b_k \cos(2\pi k t / T)] + \varepsilon_t$$

$\hat{y}_t^{(0)}$ is this fitted curve evaluated at each post- $t=0$ timestamp. In this project's own backtests it was clearly among the weakest methods whenever the underlying data had a day-level *regime shift* rather than a smooth trend — the single global β term gets misled by extrapolating a trend through what was really a level change.

STL (Seasonal-Trend decomposition via Loess). Iteratively separates the series into three additive components,

$$y_t = T_t + S_t + R_t$$

each estimated by repeated local regression (LOESS — weighted least-squares fit within a moving window, robust to outliers by down-weighting large residuals between passes). $\hat{y}_t^{(0)}$ extrapolates T_t forward and repeats the already-estimated seasonal cycle S_t . Like Fourier+trend, a linear trend extrapolation through a level shift is STL's main failure mode on this project's own test data.

Chapter 29. Box-Jenkins: ARIMA and SARIMA

Introduce the **backshift operator** B, defined by $B y_t = y_{t-1}$. An ARIMA(p, d, q) model says: after differencing the series d times to remove any trend ($\nabla^d y_t = (1-B)^d y_t$), the result follows an autoregressive-moving-average process,

$$\phi(B) (1-B)^d y_t = \theta(B) \varepsilon_t$$

$$\text{where } \phi(B) = 1 - \phi_1 B - \dots - \phi_p B^p \text{ and } \theta(B) = 1 + \theta_1 B + \dots + \theta_q B^q$$

— today's (differenced) value is a linear combination of its own past p values plus a linear combination of the past q forecast errors. `propagation.py` deseasonalises first (subtracting Chapter 27's seasonal-naïve lookup) and fits ARIMA to the residual, then adds the seasonal lookup back onto the forecast. Orders (p,d,q) are chosen by a grid search minimising the **Akaike Information Criterion**,

$$AIC = -2 \ln(L) + 2k$$

where L is the model's maximised likelihood and k its number of parameters. **SARIMA** extends this with a second, seasonal set of backshift polynomials at period m (here

m=24, hourly):

$$\phi(B) \Phi(B^m) (1-B)^d (1-B^m)^D y_t = \theta(B) \Theta(B^m) \varepsilon_t$$

SARIMA's own AIC grid search additionally covers the seasonal orders (P,D,Q), which is why it is by far the slowest single method here — and, in this project's own backtests, also the single most accurate one, at roughly ten to twenty times the runtime of everything else.

Worked illustration of order selection. Three candidate specifications fit to the same series, each with its own maximised log-likelihood $\ln(L)$ and parameter count k:

Candidate	$\ln(L)$	k	AIC = $-2\ln(L)+2k$
ARIMA(1,0,0)	-410.2	2	824.4
ARIMA(2,0,1)	-403.8	4	815.6
ARIMA(1,1,1)	-406.5	3	819.0

ARIMA(2,0,1) wins despite using more parameters than ARIMA(1,0,0): its much better log-likelihood (-403.8 vs -410.2) more than compensates for AIC's 2k penalty on the extra two parameters. ARIMA(1,1,1) sits in between — a caution against reading AIC as simply "fewer parameters is better": it penalises complexity, but only relative to how much that complexity actually improves the fit.

Chapter 30. State-Space Models: Structural Time Series and TBATS

SARIMA models only *one* seasonal period at a time and bolts a second one on afterward via a separate lookup. **State-space ("structural") models** instead represent the series as a sum of latent, separately-evolving components observed with noise:

$$y_t = \mu_t + \gamma_t + \varepsilon_t \text{ (observation equation)}$$

$$\mu_t = \mu_{t-1} + \beta_{t-1} + \eta_t \text{ (local level + trend)}$$

$$\gamma_t = -\sum_{k=1}^{m-1} \gamma_{t-k} + \omega_t \text{ (seasonal recursion, period } m)$$

where μ_t is a stochastic level, β_t a stochastic trend slope, and γ_t a stochastic seasonal component allowed to drift slowly rather than repeat identically forever, unlike Fourier's fixed-amplitude harmonics. The state disturbances are estimated jointly by the **Kalman filter** — a recursive predict-then-correct algorithm: at each step it predicts the next state

from the current one, then corrects using the newly observed y_t , weighted by each source's estimated uncertainty. A second seasonal component (weekly) is added once at least two full weekly cycles of pre-period exist.

TBATS (Box-Cox, ARMA errors, Trend, multiple Seasonal components) is purpose-built for exactly this multi-seasonal shape: it applies a Box-Cox power transform $y_t^{(\lambda)}$ to stabilise variance, models trend and several seasonal periods in the same state-space framework above, and lets the observation noise itself follow a short ARMA process rather than being pure white noise. Both fit on hourly-aggregated data and expand back to 15-minute resolution via each hour's own average within-hour deviation.

Chapter 31. The Theta Method

A top performer in the M3 and M4 forecasting competitions, specifically strong on *short* series — the regime where ARIMA's order search and the lag-feature methods of Chapter 33 both struggle for lack of data. The idea: decompose the (deseasonalised) series into two “theta-lines,” each a transform of the second differences $\nabla^2 y_t = y_t - 2y_{t-1} + y_{t-2}$ scaled by a parameter θ :

Theta-line(θ)_t has the same mean and slope as y_t , but its local curvature ∇^2 is scaled by θ

At $\theta=0$, curvature vanishes and the theta-line reduces to the ordinary least-squares straight line through the data (Chapter 5, $x=\text{time}$). At $\theta=2$, curvature is *doubled*, which empirically makes Simple Exponential Smoothing ($\hat{y}_{t+1} = \alpha y_t + (1-\alpha)\hat{y}_t$) track short-run dynamics unusually well. The final forecast combines both lines before reseasonalising.

Chapter 32. Ridge Regression in Closed Form

Three methods (this one, LightGBM, and CatBoost in Chapter 33) share one feature set: for each timestamp, how much the target was 1, 2, and 7 days prior, plus their mean, plus cyclical hour/weekday indicators. These lag features are correlated with each other by construction — exactly Chapter 4's near-collinearity concern, this time among the *features* of an ITS model rather than among the *regressors* of a hypothesis test — making plain OLS numerically unstable. **Ridge regression** penalises large coefficients, minimising

$$\beta_{\text{ridge}} = \underset{\beta}{\operatorname{argmin}} ||y - X\beta||^2 + \lambda ||\beta||^2$$

$$\beta_{\text{ridge}} = (X^T X + \lambda I)^{-1} X^T y$$

Adding λI to $X^T X$ before inverting is exactly what restores numerical stability among correlated regressors, at the cost of shrinking every coefficient slightly toward zero — the classic bias-variance tradeoff, the same tension Chapter 7's BLUE property sits inside (ridge is deliberately *not* unbiased, trading a small bias for a large variance reduction when features are this correlated). Missing lag values are mean-imputed from training-set-only statistics; if an entire feature column is NaN throughout the pre-period, its fill value is set to zero rather than the undefined mean of nothing, so the ridge equations fit a coefficient of exactly zero for it — equivalent to dropping the feature, rather than poisoning every downstream prediction with NaN.

Worked illustration of the instability ridge fixes. Four training points, two correlated features (lag1d, lag7d move almost together):

Plain OLS: $\beta = (0.132, +1.194, -0.306)$ [intercept, lag1d, lag7d]

Ridge ($\lambda=1$): $\beta = (0.375, +0.374, +0.285)$

OLS assigns lag1d a large positive coefficient and lag7d a *negative* one that nearly cancels it — a classic symptom of near-collinearity (Chapter 4): with only four points and two features that move almost together, OLS can fit the training data by pushing two coefficients to opposite extremes that happen to cancel, a solution that is highly unstable to the next data point and generalises poorly. Ridge's two coefficients instead land close to each other and both positive, a far more plausible, stable split of credit between two features carrying nearly the same information — exactly the bias-variance trade this chapter's formula was built to make.

Chapter 33. Gradient-Boosted Trees: LightGBM and CatBoost

Where Ridge fits one global linear function, gradient boosting builds an *additive ensemble of decision trees*, each correcting the previous ensemble's mistakes. Start from a constant baseline $F_0(x)$ minimising the loss over the training data, then iterate: at stage m , fit a small regression tree $h_m(x)$ to the **negative gradient** of the loss with respect to current predictions — for squared-error loss, this gradient is simply the current residual,

$$r_{i:m} = y_i - F_{m-1}(x_i)$$

$$F_m(x) = F_{m-1}(x) + \eta \cdot h_m(x)$$

where η (the learning rate) shrinks each tree's contribution so no single tree dominates and the ensemble generalises. LightGBM buckets continuous features into discrete histogram bins before searching for splits (fast, approximate); CatBoost uses an **ordered** boosting scheme specifically designed to reduce the subtle overfitting-via-leakage risk of a tree seeing a target-derived statistic computed using its own label. Both handle the lag features' NaN-heavy early rows natively. A small hyperparameter grid (tree depth, η , number of trees) is tuned for all three lag-feature methods via a held-out slice carved from the pre-period itself — never from during/post data, so tuning cannot introduce a leak either.

Chapter 34. Two Physics-Informed Methods, Mathematically

Some target columns are not independent physical quantities — they are **derived** from other columns already in the data, by a known formula. A model that respects that formula by construction should beat one that has to rediscover it statistically from a short pre-period.

RAM identity. Recall Chapter 3's identity. Rather than fitting $Y_{(0)}_t$ directly on the RAM column, this method fits a **separate** counterfactual $X_{k,t}$ for each of the seven components — sub-method matched to that term's own dynamics — and computes the projection as the exact same signed sum:

$$Y_{(0)}_t^{RAM} = \sum_{k=1}^7 s_k \cdot X_{k,t}$$

verified to floating-point precision, falling back to the ensemble (Chapter 35) entirely if any component is missing.

PTDF-flow reconstruction. Chapter 2's second formula is **not** an exact identity here, so this method validates itself before trusting its own physics, computing the ordinary **Pearson correlation coefficient** between reconstruction and actual flow on pre-period data only:

$$r = \frac{\sum_t (x_t - \bar{x})(y_t - \bar{y})}{\sqrt{[\sum_t (x_t - \bar{x})^2 \cdot \sum_t (y_t - \bar{y})^2]}}$$

with x_t the reconstruction and y_t the actual flow — exactly Chapter 5's β formula in correlation form, the two being related by $r = \beta \times (\sigma_x / \sigma_y)$. Only if r clears a minimum threshold (0.5 by default) does the method project each zone's PTDF and net position forward and combine them; otherwise it falls back to the ensemble, always recording which path was taken.

Both methods are deliberately excluded from the ensemble's own candidate pool: both fall back to the ensemble when their physics doesn't apply, and if either were also an ensemble member, that fallback would call straight back into the ensemble's own evaluation loop — unbounded recursion the moment either one's fallback condition is met.

Chapter 35. The Adaptive Ensemble, Mathematically

The ensemble backtests every *other* method on a held-out slice carved from the pre-period itself, computing each candidate method m 's **mean absolute error**,

$$MAE_m = (1/n) \sum_{t \in \text{holdout}} |y_t - Y_{\blacksquare}(0)_{m,t}|$$

Candidates whose backtest predictions are numerically identical to another's are de-duplicated first (compared via predictions themselves, not just tied MAE), keeping only the best-MAE representative. Any survivor backtesting more than 1.5× worse than the single best one is discarded. The remaining survivors are blended, weighted by the **inverse square** of their own backtest error:

$$w_m \propto 1 / MAE_m^2, Y_{\blacksquare}(0)_t = \sum_m w_m Y_{\blacksquare}(0)_{m,t} / \sum_m w_m$$

(inverse error, without squaring, measured too flat in practice). Below fourteen days of pre-period, too short to carve a reliable backtest split, the ensemble instead takes an unweighted **median** across every member rather than a mean — a plain mean measured far worse in development, since one badly-wrong member can drag a mean substantially off, while the median resists a single outlying member.

Worked illustration. Five backtested candidates, one clearly poor:

Method	MAE	Kept?	Weight $w_m \propto 1/MAE^2$
ridge	2.00	yes	0.350
theta	2.05	yes	0.333
seasonal_naive	2.10	yes	0.317
fourier_trend	3.40	no (>1.5× best)	0.000
stl	5.10	no (>1.5× best)	0.000

The best member's MAE is 2.00; 1.5× that is 3.00, so `fourier_trend` (3.40) and `stl` (5.10) are both discarded before any weight is computed, and the three survivors' weights (each

$\propto 1/\text{MAE}^2$, renormalised to sum to 1) come out nearly — but not exactly — equal, since their backtest errors were themselves close together: the squaring in Chapter 35's weighting formula rewards ridge's small edge over seasonal_naive (0.350 vs 0.317) more than a plain inverse-MAE weighting would, without letting it dominate outright when the gap between survivors is this narrow.

Part VI — Touring the Tab, Sub-Tab by Sub-Tab

You now have both the vocabulary and the mathematics. This part walks the eight sub-tabs of Maintenance Analysis in the order a first-time user would actually click through them.

Chapter 36. Setup and Outage Sources

Setup is where a session begins: load whatever market data is already on the application's Fetch/Upload tab, or generate a synthetic ninety-day demonstration dataset on the spot. Setup carries its own UTC/CET timestamp-zone toggle, independent of every other tab's, so a different timestamp assumption can be tried for **this analysis** without disturbing anything already fetched elsewhere. Apply Setup always clears every downstream result first.

Outage Sources brings in the events Parts III–V's models are tested against: a hand-curated CSV or a live fetch from the European transmission-outage transparency platform, automatically de-duplicated against whatever is already loaded.

Chapter 37. Run Analysis and Reading Results

One button runs the full battery from Part III across **every** CNEC in the loaded dataset, never just the one selected elsewhere: entity fixed effects (Chapter 9) need at least two entities to be estimable at all. The selected CNEC is used only as a report label.

Results shows one row per hypothesis, colour-coded by verdict, with a detail pane showing the full coefficient table for whichever row is selected — Chapter 23's statistical-vs-economic-significance distinction is preserved directly in how each row is labelled.

Chapter 38. Plots

Four chart types — a time series of the core parameters, a PTDF distribution, a shadow-price-versus-RAM scatter, and a CNEC binding-frequency bar chart — all drawn from the same population-level data Run Analysis just processed.

Chapter 39. Single Event: ITS, ΔRAM Decomposition, DiD, and the Per-CNEC Table

Pick one outage, a pre-period length, a post-event length, and one of the fifteen counterfactual methods from Part V — one function call then produces four views of the same event.

The **ITS pane** plots y_t against $Y_{(0)_t}$ (Chapter 26), shading the outage window. The **ΔRAM decomposition pane** applies Chapter 3's identity retrospectively. The **DiD pane** compares CNECs with historically high |PTDF| sensitivity against low-sensitivity ones, before versus during the outage:

$$ATT = (high_{during} - high_{pre}) - (low_{during} - low_{pre})$$

a genuinely different kind of evidence than a simple before/after gap, since it nets out any shock hitting high- and low-sensitivity CNECs equally. The **per-CNEC table** lists every affected CNEC's own pre, during, and Δ values side by side.

Chapter 40. The Price Spread Pane, Mathematically

Isolates one CNEC's own change in contribution to a target zone's price. A CNEC's shadow price times a zone's PTDF sensitivity on that CNEC is that CNEC's own contribution to that zone's price; comparing the **actual** value against its **counterfactual** isolates the outage's own effect:

$$impact = (SP_{actual} \times PTDF_{actual}) - (SP_{cf} \times PTDF_{cf})$$

Both factors get their own counterfactual, since PTDF itself can move during an AC-line outage (H2, Chapter 18). Expanding algebraically ($\Delta SP = SP_{actual} - SP_{cf}$, $\Delta PTDF = PTDF_{actual} - PTDF_{cf}$):

$$impact = PTDF_{cf} \cdot \Delta SP + SP_{cf} \cdot \Delta PTDF + \Delta SP \cdot \Delta PTDF$$

a price effect, a sensitivity effect, and a cross term. This is deliberately **one CNEC's own effect**, not the full zonal price change, and reuses the outage's own start time as the pre-period cutoff and whichever counterfactual method is already selected above.

Chapter 41. Explain and Export

Explain turns the Results verdicts into a plain-language paragraph per hypothesis.

Export writes the covariates, outages, and verdicts to CSV, and can produce a

self-contained HTML report embedding whatever charts are currently drawn.

Part VII — Maintaining This Tab

Chapter 42. State and Data Flow

Six pieces of state hold everything this tab knows, and every downstream sub-tab reads from these and nothing else.

State	Populated by
Loaded market data	Setup → Apply Setup
Outage events	Outage Sources (manual load or live fetch)
Covariates	Run Analysis
Regression results	Run Analysis
Hypothesis verdicts	Run Analysis
Single-event result	Single Event → Analyse Event

Chapter 43. How This Differs From the Main Application

This tab and the main six-tab dashboard both ultimately call the same underlying analytical functions and should agree on every number for identical input — but the two paths to get there are **duplicated code, not shared code**. A fix made in one path does not automatically fix the other.

Chapter 44. Known Gotchas

- Run Analysis always regresses across every CNEC in the loaded dataset, never just the one selected elsewhere — expected (Chapter 37), but the most common source of “why didn’t changing the CNEC change the result” confusion.
- The Explain sub-tab’s generated methodology text is a fixed string that always describes the seasonal-naive baseline, regardless of which of the fifteen counterfactual methods was actually used in Single Event — a known, not-yet-fixed inaccuracy.
- The exported HTML report is a separate implementation from the rest of the project’s own report generator, and does not currently include the Δ RAM decomposition chart or the DiD estimates, even though both are visible on screen.

- Editing this file's source code while a test suite is still running against it can produce a spurious test failure purely from line numbers shifting mid-run — not a real regression; re-run against a stable file to confirm.

Chapter 45. A Maintenance Checklist

When something in this tab looks wrong, check in this order:

- Is the right piece of state (Chapter 42) actually populated?
- Did Run Analysis use every CNEC, as expected — or did you expect it to filter, incorrectly?
- Is the discrepancy against the main dashboard? If so, check both orchestration paths independently (Chapter 43).
- Is it the Explain tab's methodology text? Known-stale (Chapter 44).
- Is content missing from an exported report but visible on screen? Check the Δ RAM decomposition chart and DiD text specifically.
- Could it be a timestamp misalignment? Check this tab's own independent UTC/CET toggle on Setup before assuming a data problem.

Part VIII — Hypothesis Testing, Deepened

Parts II and III built the machinery and applied it once to each hypothesis. This part goes back and finishes the job properly: the asymptotic theory that justifies treating a t-statistic as approximately normal in the first place, the general Wald-test framework every single-coefficient test in this book is secretly an instance of, worked numeric examples for the three hypotheses (H2, H4, H6) Part III stated but did not fully compute by hand, and a side-by-side comparison of all six verdicts on one consolidated report.

Chapter 46. Asymptotic Properties: Consistency and Asymptotic Normality

Chapter 7's Gauss–Markov theorem is an exact, finite-sample result — it holds for any n , however small — but it required homoskedastic, independent errors, assumptions Chapters 9 and 13 already showed this data violates. What justifies treating β as approximately normally distributed (needed for the t-statistics and p-values in Chapters 14–16) under the weaker assumptions actually used in practice is a different, **asymptotic** (large-sample) argument, resting on two properties:

- **Consistency.** As $n \rightarrow \infty$, $\hat{\beta}$ converges in probability to the true β — with enough data, the estimate gets arbitrarily close to the truth with arbitrarily high probability, even if it is not exactly unbiased at any finite n (which, e.g., Ridge in Chapter 32 is not, by deliberate design).
- **Asymptotic normality.** $\sqrt{n}(\hat{\beta} - \beta)$ converges in distribution to a normal distribution, a consequence of the Central Limit Theorem applied to the sum of (weighted) error terms inside $\hat{\beta}$'s own formula — even when the underlying ε are themselves not normally distributed, their properly-scaled sum is, approximately, once enough of them are averaged together.

Together these are why a t-statistic computed from Chapter 13's clustered sandwich variance can still be compared to a normal (or t-) reference distribution to get a valid p-value, even though the clean, finite-sample Gauss–Markov story no longer literally applies once clustering, robust variances, or (in H5's case) maximum likelihood are in play. The practical corollary is Chapter 15's power discussion made precise: these asymptotic guarantees are large- G^* , large- n^* results (technically, for clustered inference, it is the number of **clusters** G that must grow, not just n — a dataset with thousands of rows but only three or four distinct clusters offers little of the asymptotic justification this chapter describes, however large n itself looks).

Chapter 47. The Wald Test: The General Framework Behind Every t-Test in This Book

Every single-coefficient test in Chapters 17–22 — "is $\beta_{\text{■}}$ different from zero" — is a special case of a more general tool. The **Wald test** for a linear restriction $R\theta=r$ on a parameter vector θ with estimated covariance $\text{Var}(\theta_{\text{■}})$ is

$$W = (R\theta_{\text{■}} - r)_{\text{■}} [R \cdot \text{Var}(\theta_{\text{■}}) \cdot R_{\text{■}}]^{-1} (R\theta_{\text{■}} - r)$$

which, under the null hypothesis, follows a chi-squared distribution with degrees of freedom equal to the number of restrictions being tested jointly. For a *single* coefficient $\beta_{\text{■}}$ tested against zero (R is then just a row-vector selecting $\beta_{\text{■}}$, $r=0$), W collapses to exactly t^2 from Chapter 14's t-statistic — the two are the same test, expressed two ways.

The reason this generalisation matters here: Chapter 16's Holm–Bonferroni procedure corrects for testing $H1$ – $H4$ as four *separate* single-coefficient Wald tests. A genuinely different design question — "do $H1$ through $H4$ hold *jointly*", as a single combined claim" — would instead use one *multi-restriction* Wald test (R selecting all four coefficients at once, r a zero vector), which asks a different thing than four separate tests corrected after the fact, and is not currently how this tab's report is built, but is the natural tool if that joint question were ever asked.

Chapter 48. H2, Worked: AC Outages and PTDF Sensitivity, With Real Numbers

Recall H2's specification from Chapter 18: $|\text{PTDF_SRC}|_{\text{it}} = \alpha_{\text{i}} + \beta_{\text{■}} \cdot \text{ACOutage}_{\text{it}} + \varepsilon_{\text{it}}$. Two CNECs, six observations each, three before an AC-line outage and three during it:

CNEC	pre 1–3		during 4–6		
A	0.050, 0.048, 0.051		0.061, 0.063, 0.059		
B	0.031, 0.029, 0.030		0.038, 0.040, 0.037		

Applying Chapter 9's within-transformation (each CNEC demeaned by its own six-observation average) and Chapter 5's OLS to the pooled, demeaned data:

$$\beta_{\text{■}} = 0.00983, SE = 0.00098, t = 10.00, p < 0.0001 \text{ (dof} = 12 - 2 - 1 = 9\text{)}$$

A clear, precisely-estimated increase in sensitivity during the AC outage on both CNECs alike — exactly the pure topology effect H2 is built to isolate, undiluted by any accompanying change in how much power happens to be flowing (which is H1's separate question, Chapter 17).

Chapter 49. H4, Worked: Forced Outages and the Shadow Price, With Real Numbers

Recall H4's specification from Chapter 20, fit only on binding rows (shadowPrice>0). Two CNECs, four binding observations each, two pre- and two during a forced outage:

CNEC	pre (binding)	during (binding)
A	5.0, 6.0	18.0, 21.0
B	3.0, 4.0	14.0, 12.0

Within-transformed and fit as before:

$$\beta_{\text{AC}} = 11.75, SE = 1.328, t = 8.85, p = 0.0003 \text{ (dof} = 8 - 2 - 1 = 5\text{)}$$

An economically large jump in the marginal value of relaxing these constraints — the shadow price rises by roughly 12 currency units per MW on the CNECs that were already binding — consistent with a forced outage tightening an already-constrained CNEC further, exactly the mechanism H4 exists to detect, and (subject to Chapter 23's separate economic-significance floor for shadow price specifically) large enough to plausibly clear it too.

Chapter 50. H6, Worked: The Placebo, Confirmed Null

Recall H6's specification from Chapter 22 — the one hypothesis this project *wants* to fail to reject. Two CNECs, six observations each, three before and three during an outage, on FRM specifically:

CNEC	pre 1–3	during 4–6
A	80.1, 79.8, 80.3	80.0, 79.9, 80.2
B	65.2, 64.9, 65.1	65.3, 64.8, 65.0

Within-transformed and fit identically to every other hypothesis in this part:

$$\beta_{\text{FRM}} = -0.033, SE = 0.113, t = -0.294, p = 0.775 \text{ (dof} = 12-2-1 = 9\text{)}$$

A coefficient indistinguishable from zero, exactly as Chapter 2's structural, slowly-recalibrated FRM should behave under an individual outage — this is what a *passing* placebo looks like: not merely "not significant" as a throwaway result, but a specific, expected null that lets every other hypothesis's verdict in the same report be trusted rather than demoted per Chapter 22's own rule.

Chapter 51. Comparing All Six Verdicts Side by Side

Consolidating this part's and Part III's worked numbers into one report-style table — the shape `summarize_hypotheses()` actually produces, read all at once rather than one hypothesis at a time:

ID	β_{FRM}	SE	t	p (corr.)	Verdict
H1	-5.33	1.67	-3.20	0.033*	see Ch.6 — illustrative n; corrected p depends on full H1–H4 set
H2	+0.0098	0.0010	10.00	<0.0001	SUPPORTED (statistically and economically)
H3	-7.38	1.58	-4.68	0.002	SUPPORTED
H4	+11.75	1.33	8.85	0.0003	SUPPORTED
H5	+2.77 (logit)	1.58	z=1.75	0.080	Not significant at illustrative n=10
H6	-0.03	0.11	-0.29	0.775	Placebo PASSES — other verdicts trusted

Reading a report this way — all six at once, with H6 checked first — is the discipline `summarize_hypotheses()` encodes: a report where H6 fails would have every SUPPORTED verdict above demoted regardless of how clean each one's own p-value looks in isolation, because a failed placebo is evidence about the *estimator*, not about any one hypothesis specifically (Chapter 22).

Part IX — Diagnostics and Robustness

Every regression in Part III rests on assumptions — Chapter 4's four, plus homoskedasticity for the classical formulas, relaxed piece by piece through Part II. This part covers how to check whether those assumptions actually held on a given fit, and what to do when they visibly didn't.

Chapter 52. Checking the Residuals: What a Well-Behaved Fit Looks Like

A regression's residuals ($\hat{u}_i = y_i - \hat{y}_i$) are the closest thing available to the unobservable ϵ_i , and most of Chapter 4's assumptions have a direct residual-based check:

- **Exogeneity / omitted structure.** Plot residuals against fitted values, and against time. A random scatter around zero is reassuring; any visible pattern (a curve, a trend, a cluster of large residuals around a specific date) suggests the model is missing something systematic — exactly Chapter 10's omitted-CNEC-identity problem, generalised to any other missing structure.
- **Homoskedasticity.** Plot squared residuals against fitted values. A roughly flat band supports Chapter 7's classical variance formula; a widening or narrowing “cone” shape is heteroskedasticity, Chapter 12's territory.
- **Normality (relevant mainly at small samples, per Chapter 46's asymptotic argument).** A histogram or quantile-quantile plot of residuals against a normal reference. Modest departures matter little once n is large enough for the Central Limit Theorem argument to apply; they matter more in small, illustrative-sized samples like this part's own worked examples.
- **Independence over time.** Chapter 53's Durbin–Watson statistic, next.

Chapter 53. Autocorrelation and the Durbin–Watson Statistic

Chapter 13 addressed correlation *across entities at the same timestamp* via clustering. A related but distinct concern is correlation *within one entity's own series over time* — today's residual predicting tomorrow's, which clustered-by-time standard errors do not, by themselves, detect or fix. The **Durbin–Watson statistic** tests for exactly this, on a series of residuals $\hat{u}_1, \dots, \hat{u}_n$ in time order:

$$DW = \frac{\sum_{t=2}^n (\hat{u}_t - \hat{u}_{t-1})^2}{\sum_{t=1}^n \hat{u}_t^2}$$

DW ranges 0–4; a value near 2 indicates no first-order autocorrelation, values pulled toward 0 indicate positive autocorrelation (consecutive residuals tend to share a sign — a systematic run above or below the fitted line), values pulled toward 4 indicate negative (alternating) autocorrelation. Two illustrative residual sequences make the contrast concrete:

Residual pattern	DW
Runs of same-sign residuals (1.2, 1.5, 1.1, 1.4, ..., -1.1, -1.4, ...)	0.36 — strong positive autocorrelation
Alternating-sign residuals (1.2, -0.8, 0.5, -1.1, ...)	3.05 — mild negative autocorrelation, much closer to 2

A DW value far from 2 on one of this tab's own fits would suggest the residuals still carry exploitable time structure the model hasn't captured — relevant background for Part V's methods too, several of which (ARIMA's own residual-whitening objective, in particular) exist specifically to drive a **forecasting** model's residuals toward exactly this $DW \approx 2$ target.

Chapter 54. Sensitivity to Pre-Period Length and Clustering Choice

Two choices a user makes on the Setup and Single Event sub-tabs (Chapters 36, 39) have a direct line back to this part's theory, and are worth checking deliberately rather than left at their defaults:

- **Pre-period length.** Chapter 15 tied statistical power directly to sample size; Chapter 46 tied the asymptotic justification for treating t-statistics as approximately normal to a similarly large-sample regime. A verdict that flips between “significant” and “not” as the pre-period window is shortened or lengthened is a genuine sign of a result sitting right at the edge of what the available data can support — worth reporting as such, not silently picking whichever window happens to produce the more interesting-looking answer.
- **Clustering mode.** Chapter 13 showed clustered SEs can be substantially larger than naive ones; the specific ratio depends on how strong the shared, same-timestamp shocks actually are in a given window. `run_panel_regression()`'s documented fallback chain (time-clustered → entity-clustered → robust → unadjusted) means the SE actually used can differ silently between two otherwise-similar runs if one dataset supports the requested mode and another doesn't — `cov_type_used/se_fallback_occurred` (Chapter 13) exist precisely so this is visible rather than hidden inside an unexplained SE difference.

Chapter 55. A Robustness Checklist

Before trusting a SUPPORTED verdict from this tab, or reporting one further:

- Did H6 (Chapter 22, Chapter 50) pass? If not, every other verdict in the same report is suspect, not just H6 itself.
- Does the verdict clear *both* Chapter 16's corrected p-value bar and Chapter 23's economic-significance floor — not merely one or the other?
- Does `se_fallback_occurred` show the requested clustering mode was actually used, or a weaker fallback (Chapter 13, Chapter 54)?
- Is the pre-period long enough for Chapter 15's power concern and Chapter 46's asymptotic justification to plausibly apply, or is this a small-sample result closer to H5's own illustrative $n=10$ (Chapter 21) than to a genuinely large panel?
- For a Single Event ITS result specifically (Part V): does the residual/impact pattern look like Chapter 52–53's diagnostics would flag — and did the chosen counterfactual method's own minimum pre-period requirement (Part V, per-method) get met, or silently fall back to a simpler method?

Part X — A Complete Worked Case Study

Chapter 56. From Raw Outage Event to Every One of Six Verdicts: A Full Numeric Walkthrough

Tying every chapter in Parts II, III, and VIII together, start to finish, on one hypothetical but internally consistent scenario: a forced HVDC outage between a source country and two of a target zone's CNECs (A and B), covering three MTUs before the outage and three during it in the illustrative panel already used throughout Chapters 6, 10, 17–22, and 48–50.

Step 1 — Raw data and covariates

The outage event (forced, HVDC, with a known start/end timestamp) is turned into a 0/1 “active” dummy aligned to every MTU for both CNECs (Chapter 25's `build_covariates()` step). Chapter 3's RAM identity is checked to balance on every row before anything else proceeds.

Step 2 — Six specifications, six fits

Each of Chapters 17–22's equations is fit independently, all via the identical machine: Chapter 9's within-transformation removes each CNEC's own baseline, Chapter 5's OLS normal equations (or, for H5, Chapter 21's maximum-likelihood logit) produce a point estimate, and Chapter 13's clustered sandwich formula (or its documented fallback) produces a standard error.

Step 3 — The consolidated report

Chapter 51's table is exactly this step's output — six rows, one specification each, ready for the next two steps rather than read in isolation.

Step 4 — Multiple-testing correction and the placebo gate

H1–H4's four p-values go through Chapter 16's Holm–Bonferroni step-down procedure jointly; separately, H6's own result (Chapter 50: $p=0.775$, clearly not rejected) is checked as a gate on the other five, per Chapter 22's rule — since it passes here, no verdict is demoted on that account.

Step 5 — Economic significance and the final label

Each surviving coefficient is checked against Chapter 23's magnitude floor for its own dependent variable. H2's +0.0098 PTDF shift and H4's +11.75 shadow-price shift are

both large relative to typical floors for those variables in addition to being highly significant, so both are labelled SUPPORTED; H5's logit result, despite a large apparent effect size, does not clear Chapter 16's significance bar at all at this illustrative sample size, so it is reported as not significant — genuinely different, and separately worth knowing, from being reported as **disproven**.

Step 6 — One event, drilled into (Single Event / Part V)

Independently of the population-level battery above, the same forced HVDC outage can be analysed on its own via Part V's counterfactual machinery: fit $Y_{it}(0)$ (Chapter 26) on the three pre-outage MTUs for each CNEC using, say, seasonal-naïve (Chapter 27), compare it against the three during-outage MTUs actually observed, and the resulting per-timestamp impact values are exactly the gap the ITS pane (Chapter 39) plots — a genuinely different, event-specific complement to Steps 1–5's population-level verdicts, not a duplicate of them: Steps 1–5 ask whether outages **of this type, in general** move a parameter; this step asks what **this specific** outage's own effect looked like.

Every number in this walkthrough traces back to a formula stated once, in one chapter, somewhere in Parts II, III, V, or VIII — that traceability, end to end, from a raw outage record to a labelled verdict, is the entire discipline this book exists to make visible.

Appendix A — Notation Reference

Symbol	Meaning
i	Entity index (a CNEC)
t	Time index (an MTU / timestamp)
g	Cluster index (Chapter 13)
y_{it}, x_{it}, D_{it}	Outcome, generic regressor, treatment/outage dummy for entity i at time t
$\alpha, \beta, \gamma, \theta, \phi$	Population parameters (intercept, slope(s), logit coefficients, ARIMA/Theta parameters)
α_i	Entity fixed effect (Chapter 9)
ε_{it}, u_{it}	Error term / residual
hat, e.g. $\hat{\beta}$	An estimated (sample) quantity
bar, e.g. $\bar{\beta}$	A sample average
tilde, e.g. $\tilde{\beta}$	A within-transformed (demeaned) quantity
X, y	Design matrix, outcome vector (matrix notation)
$SE(\cdot)$	Standard error of an estimator
n, m, G, k	Sample size; number of hypotheses/harmonics; number of clusters; number of parameters
σ^2	Error variance
λ	Ridge penalty strength (Chapter 32)
η	Gradient-boosting learning rate (Chapter 33)
B	Time-series backshift operator, $B y_t = y_{t-1}$ (Chapter 29)
$Y_{\square(0)_t}$	Counterfactual projection at time t (Chapter 26)
s_k	Sign of the k -th RAM-identity term (Chapter 3)

Appendix B — Formula Quick Reference

Estimation (Part II)

$$\beta = (X'X)^{-1}X'y \text{ [OLS, Ch. 5]}$$

$$y_{it} = y_{it} - \alpha_i \text{ [within transform, Ch. 9]}$$

$$\text{Var_cluster}(\beta) = (X'X)^{-1}[\sum_g X_g u_{gu} X_g](X'X)^{-1} \text{ [Ch. 13]}$$

$$t = \beta/SE(\beta), p_{-}(k) \leq \alpha/(m-k+1) \text{ [Ch. 14, 16]}$$

The Six Hypotheses (Part III)

$$H1: fall_{it} = \alpha_i + \beta \cdot HVDCOutage_{it} + \epsilon_{it}$$

$$H2: |PTDF_SRC|_{it} = \alpha_i + \beta \cdot ACOutage_{it} + \epsilon_{it}$$

$$H3: RAM_{it} = \alpha_i + \beta \cdot ForcedOutage_{it} + \epsilon_{it}$$

$$H4: shadowPrice_{it} = \alpha_i + \beta \cdot ForcedOutage_{it} + \epsilon_{it} \text{ (binding rows only)}$$

$$H5: \ln(p_{it}/(1-p_{it})) = \gamma + \gamma \cdot Forced_{it}$$

$$H6: FRM_{it} = \alpha_i + \beta \cdot Outage_{it} + \epsilon_{it} \text{ (placebo)}$$

Counterfactual Methods (Part VI)

$$RAM = \sum_k s_k X_k \text{ [Ch. 3, 34]}$$

$$impact = (SP_a \times PTDF_a) - (SP_{cf} \times PTDF_{cf}) \text{ [Ch. 40]}$$

Appendix C — Glossary

Statistics and Econometrics

Term	Definition
AIC	Akaike Information Criterion, $-2\ln(L) + 2k$ — a fit-vs-complexity model-selection score (Ch. 29)
Asymptotic	Describing a property that holds as sample size grows without bound, not necessarily at any finite n (Ch. 46)
BLUE	Best Linear Unbiased Estimator (Gauss–Markov theorem, Ch. 7)
Confidence interval	A range of parameter values consistent with the data at a stated confidence level (Ch. 14)
Consistency	An estimator converging in probability to the true parameter as $n \rightarrow \infty$ (Ch. 46)
Degrees of freedom (df)	Roughly, the number of independent pieces of information left after estimating a model's parameters; divides SSE to get an unbiased σ^2 (Ch. 6)
Durbin–Watson statistic	A test for first-order autocorrelation in regression residuals (Ch. 53)
Fixed effects	Entity-specific intercepts that absorb time-invariant differences between panel members (Ch. 9)
FWER	Family-wise error rate — probability of at least one false positive across multiple tests (Ch. 16)
FWL theorem	Frisch–Waugh–Lovell — demeaning/residualising equals “controlling for” (Ch. 11)
Heteroskedasticity	Error variance that differs across observations (Ch. 12)
Holm–Bonferroni	A step-down multiple-testing correction, more powerful than plain Bonferroni (Ch. 16)
LSDV	Least Squares Dummy Variables — the (equivalent, costlier) alternative to the within estimator (Ch. 9)
MLE	Maximum-likelihood estimation — choosing parameters to maximise the probability of the observed data (Ch. 21)
Normal equations	The system $\partial S / \partial \beta = 0$ whose solution is the OLS estimator (Ch. 5)
p-value	The probability, under the null hypothesis, of a test statistic at least as extreme as the one observed (Ch. 14)
Panel data	Repeated observations of the same cross-sectional units over time (Ch. 9)

Term	Definition
Power	Probability of correctly detecting a real effect (Ch. 15)
Robust (White) SE	A standard error valid under heteroskedasticity of unknown form (Ch. 12)
Sandwich estimator	A variance formula of the form $(X'X)^{-1}[\dots](X'X)^{-1}$, used for both robust and clustered SEs (Ch. 12, 13)
t-statistic	$\beta/SE(\beta)$, the standardised distance of an estimate from a hypothesised value (Ch. 14)
Type I / II error	False positive / false negative (Ch. 15)
Wald test	The general test-statistic framework a single-coefficient t-test is a special case of (Ch. 47)
Within estimator	Fixed-effects regression via demeaning (Ch. 9)

Time Series and Machine Learning

Term	Definition
ARIMA	AutoRegressive Integrated Moving Average — a Box-Jenkins time-series model (Ch. 29)
Backshift operator (B)	$B y_t = y_{t-1}$, notation for writing time-series models compactly (Ch. 29)
Box-Cox transform	A power transform y^λ used to stabilise variance before modeling (Ch. 30)
Ensemble	A blend of several models' predictions, here weighted by inverse-squared backtest error (Ch. 35)
Gradient boosting	Building an ensemble of trees, each fit to the previous ensemble's residual/gradient (Ch. 33)
Kalman filter	A recursive predict-correct algorithm for estimating a state-space model's latent states (Ch. 30)
Learning rate (η)	The shrinkage factor applied to each new tree in gradient boosting (Ch. 33)
LOESS	Locally weighted regression, the smoothing technique behind STL (Ch. 28)
Pearson correlation (r)	A standardised measure of linear association between two series (Ch. 34)
Ridge regression	Penalised least squares, $(X'X + \lambda I)^{-1}X'y$, for collinear features (Ch. 32)
SARIMA	Seasonal ARIMA — ARIMA with an added seasonal backshift polynomial (Ch. 29)

Term	Definition
State-space model	A model representing a series as latent, evolving components observed with noise (Ch. 30)
TBATS	Box-Cox, ARMA errors, Trend, multiple Seasonal components — a state-space variant for complex seasonality (Ch. 30)
Theta method	A short-series forecasting method blending a trend line and a curvature-scaled SES line (Ch. 31)

Domain (FBMC)

Term	Definition
AMR	Adjustment for Minimum RAM (Ch. 2)
CNEC	Critical Network Element with Contingency (Ch. 2)
FAAC / AAC	Already Allocated Capacity (Ch. 2)
fall / F_allReference	The CNEC reference flow entering the RAM formula (Ch. 2)
FBMC	Flow-Based Market Coupling (Ch. 1)
fnrao / FRA	Flow from Non-costly Remedial Actions and Other adjustments (Ch. 2)
FRM	Flow Reliability Margin (Ch. 2)
IVA	Individual Validation Adjustment (Ch. 2)
MTU	Market Time Unit, 15 minutes (Ch. 2)
PTDF	Power Transfer Distribution Factor (Ch. 2)
RAM	Remaining Available Margin (Ch. 3)
Shadow price	The marginal value of relaxing a binding CNEC constraint (Ch. 2)

Appendix D — Further Reading

This book derives everything it uses, but none of it was invented here. For a reader who wants the primary or textbook sources behind each tool:

Topic	Where to go deeper
OLS, fixed effects, clustering	Wooldridge, <i>*Introductory Econometrics*</i> (panel-data chapters); Angrist & Pischke, <i>*Mostly Harmless Econometrics*</i>
Gauss–Markov, asymptotics, the Wald test	Greene, <i>*Econometric Analysis*</i>
Multiple testing (Holm–Bonferroni)	Holm (1979), “A Simple Sequentially Rejective Multiple Test Procedure,” <i>*Scandinavian Journal of Statistics*</i>
ARIMA / SARIMA (Box–Jenkins)	Box, Jenkins, Reinsel & Ljung, <i>*Time Series Analysis: Forecasting and Control*</i>
STL decomposition	Cleveland, Cleveland, McRae & Terpenning (1990), “STL: A Seasonal-Trend Decomposition Procedure Based on Loess”
State-space models and the Kalman filter	Durbin & Koopman, <i>*Time Series Analysis by State Space Methods*</i>
TBATS	De Livera, Hyndman & Snyder (2011), “Forecasting Time Series With Complex Seasonal Patterns Using Exponential Smoothing”
The Theta method	Assimakopoulos & Nikolopoulos (2000), “The Theta Model: A Decomposition Approach to Forecasting”
Ridge regression	Hoerl & Kennard (1970), “Ridge Regression: Biased Estimation for Nonorthogonal Problems”
Gradient boosting	Friedman (2001), “Greedy Function Approximation: A Gradient Boosting Machine”
FBMC methodology	The relevant TSOs' own Nordic (or Core) Flow-Based Market Coupling methodology documentation, published by the region's NEMOs/TSOs

— End —